Tugas 1

DATE :

A 1. $\frac{dy}{dx} = 3x + \sin x$

$y = \int 3x + \sin x \, dx$

$y = \frac{3}{2}x^2 - \cos x + C$ // (PU)

2. $x^2 \frac{dy}{dx} = 6 - x^3$

$dy = \frac{6}{x^2} - x \, dx$

$y = \int \frac{6}{x^2} - x \, dx$

$y = -6x^{-1} - \frac{1}{2}x^2 + C$ // (PU)

5. $\frac{dy}{dx} + x = 3$; $x = 2$, $y = 3$

$dy = 3 - x \, dx$

$y = \int 3 - x \, dx$

$y = 3x - \frac{1}{2}x^2 + C$

(PU)

$3 = 3(2) - \frac{1}{2}(2)^2 + C$

$C = 3 - 6 + 2$

$C = -1$

$y = 3x - \frac{1}{2}x^2 - 1$ //

(PK)

B 1. $\frac{dv}{dt} + kv^2 = 0$, kecepatan saat $t = 0$ dan $v = v_0$ (konstan)

$\frac{dv}{dt} = -kv^2$

$\int \frac{1}{-kv^2} dv = \int dt$

$\frac{-1}{kv} = t + C$

$v = -\frac{1}{k(t+C)}$

$t = 0$ & $v = v_0$

$v_0 = -\frac{1}{kC} \rightarrow C = -\frac{1}{kv_0}$

$\therefore v = -\frac{1}{k(t - \frac{1}{kv_0})}$ //

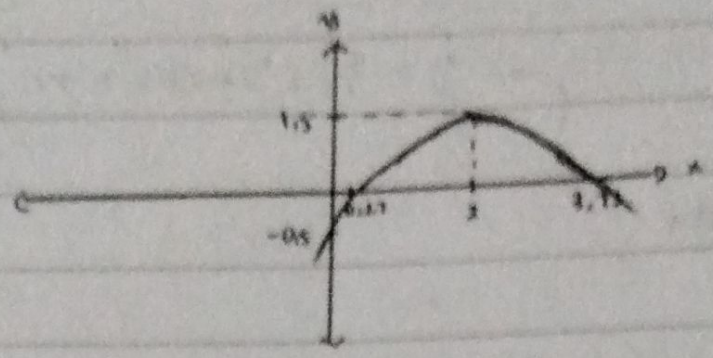
2. $\frac{dy}{dx} = 2 - x \rightarrow y = \int 2 - x \, dx$

$y = 2x - \frac{1}{2}x^2 + C$

$1 = 2(1) - \frac{1}{2}(1)^2 + C$

$C = -\frac{1}{2}$

$y = 2x - \frac{1}{2}x^2 - \frac{1}{2}$



3. Rangkaian seri dgn $R = 25\ \Omega$ dan $L = 10$ H dengan $E(t) = 100$, jika saat $t=0$ maka $i = 0$, $i(t) = ?$ dan $q(t) > 0$

$$10\frac{di}{dt} + 25i = 100$$

→ $P = 2{,}5$ dan $Q = 10$

faktor integral $e^{\int 2{,}5\,dt} = e^{2{,}5t}$

$$\frac{di}{dt} + 2{,}5i = 10$$

$$i(t)\,e^{2{,}5t} = \int 10\,e^{2{,}5t}\,dt$$

$$i(t)\,e^{2{,}5t} = 4e^{2{,}5t} + C$$

$$i(t) = 4 + Ce^{2{,}5t}$$

→ $i(0) = 4 + Ce^{2{,}5(0)}$

$$0 = 4 + C$$

$$C = -4$$

→ $i(t) = 4 - 4e^{2{,}5t}$

$$i(t) = 4(1 - e^{2{,}5t})$$ //

$$q(t) = \int i(t)\,dt$$

$$= \int 4(1 - e^{2{,}5t})\,dt$$

$$= 4\left(t - \frac{2e^{2{,}5t}}{5}\right) + C$$

→ $q(0) = 4\left(0 - \frac{2e^{2{,}5(0)}}{5}\right) + C$

$$0 = 4\left(0 - \frac{2}{5}\right) + C$$

$$C = \frac{8}{5}$$

$$q(t) = 4\left(t - \frac{2e^{2{,}5t}}{5}\right) + \frac{8}{5}$$ //

DATE :

C 1. $\frac{dy}{dx} - y = x$ faktor integral e^{-x}

$u = x \quad dv = e^{-x}$

$du = dx \quad v = -e^{-x}$

$e^{-x} y = \int x e^{-x} dx$

$\frac{1}{e^x} y = \frac{-x}{e^x} - \left(\frac{1}{e^x}\right) + C$

$y = -x - 1 + Ce^x$

2. $\frac{dy}{dx} - 2y = 8e^x$ faktor integral e^{-2x}

$\frac{1}{e^{2x}} y = \int 8e^x \cdot \frac{1}{e^{2x}} dx$

$\frac{1}{e^{2x}} y = \frac{-8}{e^x} + C$

$y = -8e^x + Ce^{2x}$

3. $\frac{dy}{dx} + 5y = 20$ faktor integral e^{5x} $y(0) = 2$

$e^{5x} y = \int 20e^{5x} dx$

$e^{5x} y = 4e^{5x} + C$

$y = 4 + \frac{C}{e^{5x}}$ (PU)

$y(0) = 4 + \frac{C}{e^{5(0)}}$

$2 = 4 + C$

$C = -2$

$y = 4 - \frac{2}{e^{5x}}$ (PK) //